

AN EXCLUSIVE CLUB

USA

In the past decade the US has invested heavily on building ground-based interceptors but still has only a rudimentary anti-ballistic missile (ABM) system in place

MISSILES: It uses a combination of THAAD, SM3 and GBI systems for strikes of 50 km and above and for its second line of defence the PAC3



REUTERS

RUSSIA

It was unhappy when the US pulled out of the ABM treaty in 2001. It has now revived its programme to build sophisticated missile defence systems.

MISSILES: Russia relies largely on endoatmospheric missiles of the S-300 and S-400 series that can strike targets at an altitude of 20 km and above



AFP

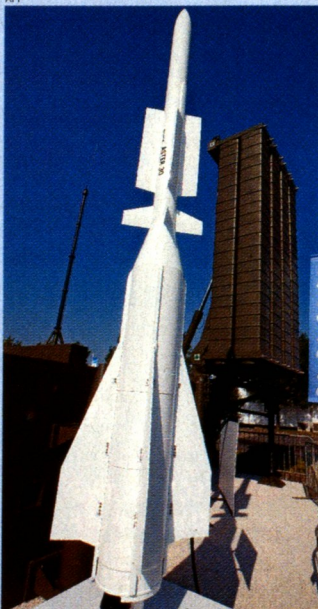
Only four other nations have anti-ballistic missile systems

FRANCE

A relative new-comer France has made steady progress with ABM systems

MISSILES: ASTER, its main interceptor, has a capability of knocking down incoming missiles at a height of 40 km

AFP



AFP



ISRAEL

Along with the US, Israel has jointly developed its ABM systems including the most sophisticated long-range radars.

MISSILES: Israel uses highly accurate ARROW systems for its line of defence.

outside the earth's atmosphere). While dealing with nuclear missiles, "you can't afford to let even one go through and strike a target", says retired Air Commodore Jasjit Singh, director of the Centre for Air Power Studies. So it calls for a multi-layered ballistic missile defence that would give field commanders "a second chance" to strike down an enemy missile as it enters the earth's atmosphere between 15 and 30 km above the earth.

Almost a year after the first successful launch, DRDO's air defence systems team in Hyderabad tasked with the ABM project was ready with endoatmospheric (within the earth's atmosphere) interceptor missiles that would give India the much-needed second layer of defence. Known as AAD01 (Advanced Air Defence 01), the interceptor had to be far more agile and faster than PAD01, its higher-flying cousin. PAD01 was a modified version of the Prithvi—it used its liquid fuel engine in the boost phase and for the second stage "kill vehicle", the team had developed a powerful solid rocket that included special divert thrusters giving it a high degree of manoeuvrability. It was also equipped with the terminal homing guidance system to carry out mid-course corrections.

For AAD01, an all-new missile had to be developed—a seven-metre high single-stage solid rocket with specially-built jet vanes to give it an even higher degree of manoeuvrability than the PAD01. "It's lean and mean", says V.K. Saraswat, DRDO's chief controller for missiles and programme director of the air defence systems. The missile also had to travel at supersonic speeds equivalent to 4.5 mach (four-and-a-half times the speed of sound or 5,512 kmph). Like PAD01 it too was equipped with a navigation system and a homing seeker that can automatically carry out adjustments with reference to the incoming enemy missile.

The first test of AAD01 was also at the Wheeler Island and the entire team moved to the launch pad in the last week of November. They first carried out a proof-of-concept test by launching an interceptor against an imaginary missile. When that succeeded, they scheduled a strike for a live target—again an incoming Prithvi—for the morning of December 6. An hour before